Solutions

Maths

Q1. Let X be a random variable following a binomial distribution with parameters n and p. If the variance of X is $\frac{24}{25}$ and the probability of success in a single trial is 0.4, then the probability that X takes a value greater than or equal to its expectation is:

(A) $\frac{328}{625}$ (B) $\frac{216}{625}$ (C) $\frac{297}{625}$ (D) $\frac{343}{625}$

Ans. A

Sol. For a binomial distribution, the variance is given by:

$$\text{Var}(X) = np(1-p)$$

Given that $\text{Var}(X) = \frac{24}{25}$ and $p = 0.4 = \frac{2}{5}$, we have $1-p = \frac{3}{5}$. Substituting these values into the formula gives:

$$n\left(\frac{2}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25} \implies \frac{6n}{25} = \frac{24}{25} \implies n = 4$$

The mathematical expectation (mean) of X is:

$$E[X] = np = 4 \times \frac{2}{5} = 1.6$$

We need to evaluate the probability $P(X \geq E[X])$, which is $P(X \geq 1.6)$. Since X can only take non-negative integer values, this is equivalent to $P(X \geq 2)$.

$$P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$$

Using the binomial probability mass function $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$, we compute:

$$P(X=0) = \binom{4}{0}\left(\frac{2}{5}\right)^0\left(\frac{3}{5}\right)^4 = \frac{81}{625}$$

$$P(X=1) = \binom{4}{1}\left(\frac{2}{5}\right)^1\left(\frac{3}{5}\right)^3 = 4 \times \frac{2}{5} \times \frac{27}{125} = \frac{216}{625}$$

Combining these terms:

$$P(X \geq 2) = 1 - \left(\frac{81}{625} + \frac{216}{625}\right) = 1 - \frac{297}{625} = \frac{328}{625}$$

Hence, the correct option is (A).

Q2. Let $x_1, x_2, \ldots, x_{20}$ be 20 real observations such that $\sum_{i=1}^{20}(x_i-2)^2 = 100$ and $\sum_{i=1}^{20}(x_i-4)^2 = 180$. If the mean and variance of the observations $y_1, y_2, \ldots, y_{20}$ given by $y_i = 3x_i + k$ (where k is a constant) are μ and σ^2 respectively, such that $\mu^2 + \sigma^2 = 46$, then the sum of all possible values of k is:

(A) -12 (B) -6 (C) 12 (D) 6

Ans. A

Sol. Expanding the given summations:

$$\sum_{i=1}^{20} x_i^2 - 4\sum_{i=1}^{20} x_i + 80 = 100 \implies \sum_{i=1}^{20} x_i^2 - 4\sum_{i=1}^{20} x_i = 20 \quad \text{--- (1)}$$

$$\sum_{i=1}^{20} x_i^2 - 8\sum_{i=1}^{20} x_i + 320 = 180 \implies \sum_{i=1}^{20} x_i^2 - 8\sum_{i=1}^{20} x_i = -140 \quad \text{--- (2)}$$

Subtracting equation (2) from equation (1):

$$4\sum_{i=1}^{20} x_i = 160 \implies \sum_{i=1}^{20} x_i = 40$$

Thus, the mean of the observations x_i is:

$$\bar{x} = \frac{40}{20} = 2$$

Substituting $\sum_{i=1}^{20} x_i = 40$ into equation (1):

$$\sum_{i=1}^{20} x_i^2 - 160 = 20 \implies \sum_{i=1}^{20} x_i^2 = 180$$

The variance of the observations x_i is:

$$\text{Var}(x) = \frac{\sum_{i=1}^{20} x_i^2}{20} - (\bar{x})^2 = \frac{180}{20} - 2^2 = 9 - 4 = 5$$

For the transformed observations $y_i = 3x_i + k$:

The new mean is:

$$\mu = 3\bar{x} + k = 6 + k$$

The new variance is:

$$\sigma^2 = 3^2 \cdot \text{Var}(x) = 9 \times 5 = 45$$

Using the given relation $\mu^2 + \sigma^2 = 46$:

$$(6+k)^2 + 45 = 46 \implies (6+k)^2 = 1$$

$$\implies 6 + k = 1 \quad \text{or} \quad 6 + k = -1$$

$$\implies k = -5 \quad \text{or} \quad k = -7$$

The sum of all possible values of k is:

$$(-5) + (-7) = -12$$

Hence, the correct option is (A).

Q3. Let a and b be positive real numbers such that

$$\lim_{x \to 0} \frac{\sin(ax) - \ln(1 + bx)}{x^2} = 2$$

If $f(x) = \tan^{-1}\left(\frac{\sqrt{1 + x^a} - 1}{x^{b-1}}\right)$ for $x > 0$, then the value of $f'(1)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) 1

Ans. B

Sol. Using expansions for $\sin(ax)$ and $\ln(1 + bx)$ in the given limit:

$$\sin(ax) = ax - \frac{a^3x^3}{6} + \mathcal{O}(x^5)$$

$$\ln(1 + bx) = bx - \frac{b^2x^2}{2} + \frac{b^3x^3}{3} - \mathcal{O}(x^4)$$

Substituting these into the limit expression gives:

$$\lim_{x \to 0} \frac{(a - b)x + \frac{b^2}{2}x^2 - \left(\frac{a^3}{6} + \frac{b^3}{3}\right)x^3 + \ldots}{x^2} = 2$$

For the limit to exist and be finite, the coefficient of x must vanish:

$$a - b = 0 \implies a = b$$

Evaluating the limit with $a = b$:

$$\frac{b^2}{2} = 2 \implies b^2 = 4 \implies b = 2 \quad (\text{since } b > 0)$$

Thus, $a = 2$ and $b = 2$.
Substituting $a = 2$ and $b = 2$ into the expression for $f(x)$:

$$f(x) = \tan^{-1}\left(\frac{\sqrt{1 + x^2} - 1}{x}\right)$$

Let $x = \tan\theta$ where $\theta \in \left(0, \frac{\pi}{2}\right)$ for $x > 0$:

$$\frac{\sqrt{1+x^2}-1}{x} = \frac{\sec\theta - 1}{\tan\theta} = \frac{1-\cos\theta}{\sin\theta} = \tan\left(\frac{\theta}{2}\right)$$

Therefore:

$$f(x) = \tan^{-1}\left(\tan\left(\frac{\theta}{2}\right)\right) = \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x$$

Differentiating $f(x)$ with respect to x:

$$f'(x) = \frac{1}{2(1+x^2)}$$

Evaluating at $x = 1$:

$$f'(1) = \frac{1}{2(1+1^2)} = \frac{1}{4}$$

Hence, the correct option is (B).

Q4. The total number of 5-digit numbers that can be formed using the digits $1, 2, 3, 4, 5, 6, 7, 8$ without repetition such that the product of any two consecutive digits is even, is

(A) 2496 (B) 2208 (C) 2016 (D) 2544

Ans. A

Sol. The condition that the product of any two consecutive digits is even means that no two odd digits can be adjacent.

The available digits consist of 4 odd digits $\{1, 3, 5, 7\}$ and 4 even digits $\{2, 4, 6, 8\}$.

Since it is a 5-digit number without repetition, we analyze based on the count of odd digits:

Case 1: 3 odd digits and 2 even digits

The digits must be in the pattern O-E-O-E-O.

Ways to select and arrange: $\binom{4}{3} \times \binom{4}{2} \times 3! \times 2! = 4 \times 6 \times 6 \times 2 = 288$.

Case 2: 2 odd digits and 3 even digits

Select 3 even digits and arrange them: $\binom{4}{3} \times 3! = 24$ ways.

This creates 4 gaps. Select 2 odd digits and arrange them in these gaps:

$\binom{4}{2} \times \binom{4}{2} \times 2! = 6 \times 6 \times 2 = 72$ ways.

Total for this case: $24 \times 72 = 1728$.

Case 3: 1 odd digit and 4 even digits

Select 4 even digits and arrange them: $\binom{4}{4} \times 4! = 24$ ways.

This creates 5 gaps. Select 1 odd digit and place it: $\binom{4}{1} \times 5 = 20$ ways.

Total for this case: $24 \times 20 = 480$.

Summing the cases:

$$288 + 1728 + 480 = 2496$$

Correct option is (A).

Q5. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \left(\lim\limits_{n \to \infty} \sum\limits_{k=1}^{n} \tan^{-1} \left(\dfrac{2x}{x^2 + 4k^2 - 1} \right) \right) + \cos^{-1} \left(\dfrac{1 - x^2}{1 + x^2} \right)$ where the inverse trigonometric functions assume their principal values. Which of the following statements is (are) **TRUE**?

(A) $f(x)$ is continuous for all $x \in \mathbb{R}$ but not differentiable at $x = 0$.

(B) $x = 0$ is a point of local minima of $f(x)$.

(C) The equation $f(x) = \dfrac{\pi}{3}$ has exactly two distinct real solutions.

(D) The equation $f(x) = \dfrac{\pi}{2}$ has exactly two distinct real solutions.

Ans. A, B, C

Sol. Let the given infinite series be $S(x) = \lim\limits_{n \to \infty} \sum\limits_{k=1}^{n} \tan^{-1} \left(\dfrac{2x}{x^2 + 4k^2 - 1} \right)$. For $x > 0$, the general term can be rewritten as:

$$\tan^{-1} \left(\frac{\frac{2k+1}{x} - \frac{2k-1}{x}}{1 + \left(\frac{2k+1}{x} \right) \left(\frac{2k-1}{x} \right)} \right) = \tan^{-1} \left(\frac{2k+1}{x} \right) - \tan^{-1} \left(\frac{2k-1}{x} \right)$$

This is a telescoping series, so the sum of the first n terms is:

$$S_n(x) = \tan^{-1} \left(\frac{2n+1}{x} \right) - \tan^{-1} \left(\frac{1}{x} \right)$$

Taking the limit as $n \to \infty$:

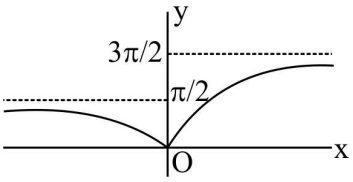
$$S(x) = \frac{\pi}{2} - \tan^{-1} \left(\frac{1}{x} \right) = \tan^{-1}(x) \quad (\text{for } x > 0)$$

Since the general term is an odd function of x and $S(0) = 0$, we deduce that $S(x) = \tan^{-1}(x)$ for all $x \in \mathbb{R}$. Next, consider the second term $g(x) = \cos^{-1} \left(\dfrac{1 - x^2}{1 + x^2} \right)$. Using standard properties:

$$g(x) = \begin{cases} 2 \tan^{-1}(x), & x \geq 0 \\ -2 \tan^{-1}(x), & x < 0 \end{cases}$$

Combining $S(x)$ and $g(x)$, we obtain the explicit form of $f(x)$:

$$f(x) = \begin{cases} 3\tan^{-1}(x), & x \geq 0 \\ -\tan^{-1}(x), & x < 0 \end{cases}$$



from graph options ABC are correct.

Q6. Let X be a discrete random variable taking values from the set $\{0, 1, 2, 3\}$. Suppose that for each $k \in \{0, 1, 2, 3\}$, the probability mass function is given by $P(X = k) = a + bk$, where a and b are real constants. If the mean (expectation) of X is 2, then which of the following statements is (are) **TRUE**?

(A) $P(X \geq 2) = \dfrac{7}{10}$ (B) The variance of X is 1 (C) $P(X \text{ is odd} \mid X \geq 1) = \dfrac{2}{3}$

(D) The variance of the random variable $2X + 3$ is 5

Ans. A, B, C

Sol. Since the sum of all probabilities in a probability distribution must be equal to 1:

$$\sum_{k=0}^{3} P(X = k) = 1 \implies (a) + (a + b) + (a + 2b) + (a + 3b) = 1$$

$$4a + 6b = 1 \quad \text{--- (1)}$$

The mean of the random variable X is given as 2:

$$E[X] = \sum_{k=0}^{3} k \cdot P(X = k) = 2 \implies 0(a) + 1(a + b) + 2(a + 2b) + 3(a + 3b) = 2$$

$$6a + 14b = 2 \implies 3a + 7b = 1 \quad \text{--- (2)}$$

Solving equations (1) and (2) simultaneously:
From (1), $2a + 3b = 0.5 \implies 6a + 9b = 1.5$
From (2), $6a + 14b = 2$
Subtracting the two rewritten equations yields $5b = 0.5 \implies b = \dfrac{1}{10}$.
Substituting $b = \dfrac{1}{10}$ into (2) gives $3a + \dfrac{7}{10} = 1 \implies a = \dfrac{1}{10}$.
Thus, the probability distribution is:

- $P(X = 0) = \dfrac{1}{10}$
- $P(X = 1) = \dfrac{2}{10}$
- $P(X = 2) = \dfrac{3}{10}$

• $P(X=3)=\dfrac{4}{10}$

Evaluating the options:

• For option (A):

$$P(X \geq 2)=P(X=2)+P(X=3)=\frac{3}{10}+\frac{4}{10}=\frac{7}{10}$$

Hence, option (A) is correct.

• For option (B):

$$E[X^2]=\sum_{k=0}^{3} k^2 \cdot P(X=k)=0^2\left(\frac{1}{10}\right)+1^2\left(\frac{2}{10}\right)+2^2\left(\frac{3}{10}\right)+3^2\left(\frac{4}{10}\right)$$

$$E[X^2]=\frac{0+2+12+36}{10}=\frac{50}{10}=5$$

$$\operatorname{Var}(X)=E[X^2]-(E[X])^2=5-2^2=1$$

Hence, option (B) is correct.

• For option (C):

$$P(X \text{ is odd} \mid X \geq 1)=\frac{P(X=1 \text{ or } X=3)}{P(X \geq 1)}=\frac{\frac{2}{10}+\frac{4}{10}}{1-P(X=0)}=\frac{\frac{6}{10}}{\frac{9}{10}}=\frac{6}{9}=\frac{2}{3}$$

Hence, option (C) is correct.

• For option (D):

$$\operatorname{Var}(2X+3)=2^2 \cdot \operatorname{Var}(X)=4 \cdot 1=4 \neq 5$$

Hence, option (D) is incorrect.

Correct Options: (A), (B), and (C)

Q7. Let $f:\mathbb{R} \to \mathbb{R}$ be a periodic function defined by:

$$f(x)=\begin{cases}\{x\}\left[\frac{1}{\{x\}}\right] & \text{if } x \notin \mathbb{Z} \\ 1 & \text{if } x \in \mathbb{Z}\end{cases}$$

where $\{x\}$ denotes the fractional part of x and $[x]$ denotes the greatest integer less than or equal to x. Consider another function $g:\mathbb{R} \to \mathbb{R}$ defined as $g(x)=\sin^2(\pi x)f(x)$. Then which of the following statements is (are) **TRUE**?

(A) $f(x)$ is continuous at all $x \in \mathbb{Z}$ (B) $f(x)$ is differentiable at all $x \in \mathbb{Z}$

(C) $g(x)$ is differentiable at all $x \in \mathbb{Z}$ and $g'(n)=0$ for all $n \in \mathbb{Z}$

(D) The set of points of discontinuity of $f(x)$ is identical to the set of points of discontinuity of $g(x)$

Ans. A, C, D

Sol. Since $f(x)$ depends solely on $\{x\}$, it is periodic with period 1. To analyze its behavior at integers, we evaluate the limits at $x = 0$: • Right-hand limit ($x \to 0^+$): Here $\{x\} = x$.

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} x\left[\frac{1}{x}\right]$$

Using the squeeze theorem with $\frac{1}{x} - 1 < \left[\frac{1}{x}\right] \leq \frac{1}{x}$, we get $\lim_{x\to 0^+} f(x) = 1$. • Left-hand limit ($x \to 0^-$): Let $x = -t$ where $t \to 0^+$. Then $\{x\} = 1 - t$. For small $t > 0$, $1 - t \in \left(\frac{1}{2}, 1\right)$, which implies $\left[\frac{1}{1-t}\right] = 1$.

$$\lim_{x\to 0^-} f(x) = \lim_{t\to 0^+} (1-t)\cdot 1 = 1$$

Since $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x) = f(0) = 1$, $f(x)$ is continuous at $x = 0$, and by periodicity, at all $x \in \mathbb{Z}$. Thus, statement (A) is true. Now, checking the differentiability of $f(x)$ at $x = 0$:

$$\text{RHD} = \lim_{x\to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x\to 0^+} \frac{x\left[\frac{1}{x}\right] - 1}{x} = \lim_{x\to 0^+} \left(\left[\frac{1}{x}\right] - \frac{1}{x}\right) = \lim_{x\to 0^+} \left(-\left\{\frac{1}{x}\right\}\right)$$

Since $\left\{\frac{1}{x}\right\}$ oscillates infinitely between 0 and 1 as $x \to 0^+$, the limit does not exist. Thus, $f(x)$ is non-differentiable at all integers. Statement (B) is false. For the function $g(x) = \sin^2(\pi x) f(x)$, we check differentiability at any integer $n \in \mathbb{Z}$:

$$g'(n) = \lim_{x\to n} \frac{\sin^2(\pi x) f(x) - 0}{x - n} = \lim_{x\to n} \left(\frac{\sin(\pi x)}{x - n}\right) \cdot \sin(\pi x) f(x)$$

Since $\lim_{x\to n} \frac{\sin(\pi x)}{x - n} = \pi(-1)^n$, $\lim_{x\to n} \sin(\pi x) = 0$, and $f(x)$ is bounded near n, the limit evaluates to 0. Thus, $g(x)$ is differentiable at all integers and $g'(n) = 0$. Statement (C) is true. The points of discontinuity for $f(x)$ occur inside the interval $(0, 1)$ where $\frac{1}{x} \in \mathbb{Z}$, which yields $x = n + \frac{1}{k}$ for $k \in \mathbb{Z}, |k| \geq 2, n \in \mathbb{Z}$. At these non-integral points, $\sin^2(\pi x) \neq 0$. Consequently, the jump discontinuities of $f(x)$ are preserved in $g(x)$. Since both functions are continuous at all integers, their sets of points of discontinuity are precisely identical. Statement (D) is true. Correct Options: (A), (C) and (D)

Q8. A box contains 9 balls, of which 3 are red, 3 are green, and 3 are blue. Three balls are drawn from the box sequentially, one by one, without replacement. Let E_1 be the event that at least one red ball is drawn in the first two draws. Let E_2 be the event that at least one green ball is drawn in the second and third draws. Let E_3 be the event that at least one blue ball is drawn in the first and third draws. Which of the following statements is (are) **TRUE**?

(A) The probability that at least one of the three events occurs is $\frac{53}{56}$.

(B) The probability that all three events occur simultaneously is $\frac{3}{28}$.

(C) The conditional probability $P(E_1|E_2)$ is equal to $\frac{17}{35}$.

(D) The probability that exactly one of the three events occurs is $\frac{1}{4}$.

Ans. A, B, D

Sol. The total number of ways to sequentially draw 3 balls out of 9 without replacement is:

$$n(S) = 9 \times 8 \times 7 = 504$$

Consider the complement event where none of the events E_1, E_2, E_3 occur, denoted as $E_1^c \cap E_2^c \cap E_3^c$.

• $E_1^c \implies$ Neither the 1st nor the 2nd ball is Red.

• $E_2^c \implies$ Neither the 2nd nor the 3rd ball is Green.

• $E_3^c \implies$ Neither the 1st nor the 3rd ball is Blue.

This strictly forces the unique ordered sequence of colors for the three draws to be Green (G) $\rightarrow$ Blue (B) $\rightarrow$ Red (R).

Using the Multiplication Theorem:

$$P(GBR) = P(G_1) \cdot P(B_2|G_1) \cdot P(R_3|G_1 \cap B_2) = \frac{3}{9} \times \frac{3}{8} \times \frac{3}{7} = \frac{27}{504} = \frac{3}{56}$$

Thus, by the Principle of Inclusion-Exclusion / Complements:

$$P(E_1 \cup E_2 \cup E_3) = 1 - P(E_1^c \cap E_2^c \cap E_3^c) = 1 - \frac{3}{56} = \frac{53}{56}$$

Hence, option (A) is TRUE.

For all three events to occur simultaneously ($E_1 \cap E_2 \cap E_3$), we analyze the required color configurations:

• If the 1st ball is Red $\implies$ 3rd must be Blue (from E_3) $\implies$ 2nd must be Green (from E_2). This gives the sequence RGB.

• If the 1st ball is not Red $\implies$ 2nd must be Red (from E_1) $\implies$ 3rd must be Green (from E_2) $\implies$ 1st must be Blue (from E_3). This gives the sequence BRG.

Number of favorable ways for each of these two sequences is $3 \times 3 \times 3 = 27$.

$$n(E_1 \cap E_2 \cap E_3) = 27 + 27 = 54$$

$$P(E_1 \cap E_2 \cap E_3) = \frac{54}{504} = \frac{6}{56} = \frac{3}{28}$$

Hence, option (B) is TRUE.

By symmetry, the individual probabilities of the events are equal:

$$P(E_1^c) = \frac{6}{9} \times \frac{5}{8} = \frac{5}{12} \implies P(E_1) = P(E_2) = P(E_3) = \frac{7}{12}$$

Using the Principle of Inclusion-Exclusion for three symmetric events:

$$P(E_1 \cup E_2 \cup E_3) = 3P(E_1) - 3P(E_1 \cap E_2) + P(E_1 \cap E_2 \cap E_3)$$

$$\frac{53}{56} = 3\left(\frac{7}{12}\right) - 3P(E_1 \cap E_2) + \frac{6}{56}$$

$$3P(E_1 \cap E_2) = \frac{7}{4} + \frac{6}{56} - \frac{53}{56} = \frac{98 + 6 - 53}{56} = \frac{51}{56} \implies P(E_1 \cap E_2) = \frac{17}{56}$$

Now, the conditional probability is:

$$P(E_1|E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{17/56}{7/12} = \frac{17}{56} \times \frac{12}{7} = \frac{51}{98}$$

Hence, option (C) is FALSE.
The probability that exactly one of the three events occurs is given by:

$$P(\text{exactly 1}) = \sum P(E_i) - 2\sum P(E_i \cap E_j) + 3P(E_1 \cap E_2 \cap E_3)$$

$$P(\text{exactly 1}) = 3\left(\frac{7}{12}\right) - 6\left(\frac{17}{56}\right) + 3\left(\frac{6}{56}\right) = \frac{7}{4} - \frac{102}{56} + \frac{18}{56}$$

$$P(\text{exactly 1}) = \frac{98 - 102 + 18}{56} = \frac{14}{56} = \frac{1}{4}$$

Hence, option (D) is TRUE.
Correct Options: A, B, D

Q9. Consider a collection of 12 letters consisting of four identical copies of 'A', four identical copies of 'B', and four identical copies of 'C'. Let N be the total number of unique linear arrangements of these 12 letters. Which of the following statements is (are) **TRUE**?

(A) The number of arrangements in which no two 'A's are adjacent is 8820.

(B) The number of arrangements that form a palindrome (read the same from left to right as from right to left) is 90.

(C) The number of arrangements in which all identical copies of at least one type of letter are together is 1890.

(D) The number of arrangements in which all 'A's are together but no two 'B's are adjacent is 75.

Ans. A, B, D

Sol. The total number of unique arrangements of the 12 letters is given by:

$$N = \frac{12!}{4!4!4!} = 34650$$

Analysis of Option (A):
To ensure no two 'A's are adjacent, we first arrange the four 'B's and four 'C's.
Number of ways to arrange 4 'B's and 4 'C's $= \frac{8!}{4!4!} = 70$ ways.
These 8 letters create $8 + 1 = 9$ available gaps. We choose 4 gaps out of 9 to place the 4 identical 'A's:

$$\text{Ways} = \binom{9}{4} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$$

Total valid ways $= 70 \times 126 = 8820$.
Hence, statement (A) is TRUE.
Analysis of Option (B):
For a linear arrangement of length 12 to be a palindrome, the first 6 positions must completely mirror the last 6 positions. Thus, the first 6 slots must contain exactly half of the total count of each letter type, i.e., 2 'A's, 2 'B's, and 2 'C's.
The number of ways to arrange these letters in the first 6 positions is:

$$\text{Ways} = \frac{6!}{2!2!2!} = \frac{720}{8} = 90$$

Each such arrangement uniquely defines the remaining 6 positions.
Hence, statement (B) is TRUE.
Analysis of Option (C):
Let X, Y, and Z represent the sets of arrangements where all 'A's are together, all 'B's are together, and all 'C's are together, respectively. We need to find $n(X \cup Y \cup Z)$ using the Principle of Inclusion-Exclusion (PIE):
• $n(X) = \frac{9!}{4!4!} = 630$ (treating the four 'A's as 1 block, along with 4 'B's and 4 'C's). By symmetry, $n(X) = n(Y) = n(Z) = 630$.
• $n(X \cap Y) = \frac{6!}{4!} = 30$ (treating 'A's as 1 block, 'B's as 1 block, along with 4 'C's). By symmetry, $n(X \cap Y) = n(Y \cap Z) = n(Z \cap X) = 30$.
• $n(X \cap Y \cap Z) = 3! = 6$ (treating each letter type as a single block).
Applying PIE:

$$n(X \cup Y \cup Z) = 3(630) - 3(30) + 6 = 1890 - 90 + 6 = 1806$$

Since $1806 \neq 1890$, statement (C) is FALSE.
Analysis of Option (D):
When all 'A's are together, they form a single block $(AAAA)$. We need to arrange this block along with 4 identical 'B's and 4 identical 'C's such that no two 'B's are adjacent.

First, we arrange the non-'B' objects, which are the block $(AAAA)$ and the 4 'C's:

$$\text{Ways} = \frac{5!}{1!4!} = 5 \text{ ways}$$

These 5 objects create $5 + 1 = 6$ gaps. We choose 4 gaps out of 6 to insert the 4 identical 'B's:

$$\text{Ways} = \binom{6}{4} = 15$$

Total valid ways $= 5 \times 15 = 75$.
Hence, statement (D) is TRUE.
Correct Options: (A), (B) and (D)

Q10. Let n be a positive integer such that $n \geq 2$. Consider the following summations involving the binomial coefficients $\binom{n}{r}$:

$$A_n = \sum_{r=0}^{n} \binom{n}{r}^2$$

$$B_n = \sum_{r=1}^{n} r\binom{n}{r}^2$$

$$C_n = \sum_{r=1}^{n} r^2\binom{n}{r}^2$$

Which of the following statements is (are) TRUE?

(A) $B_n = \frac{n}{2}A_n$ (B) $C_n = n^2\binom{2n-2}{n-1}$ (C) $\frac{C_n}{B_n} = \frac{n^2}{2n-1}$ (D) For $n = 3$, $A_3 + B_3 = 60$

Ans. A, B, C

Sol. Correct Options: A, B, C
• Analysis for A_n and B_n:
Using the standard identity $\binom{n}{r} = \binom{n}{n-r}$, we have:

$$A_n = \sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r} = \binom{2n}{n}$$

For B_n, substituting r with $n - r$ yields:

$$B_n = \sum_{r=0}^{n}(n-r)\binom{n}{n-r}^2 = n\sum_{r=0}^{n}\binom{n}{r}^2 - \sum_{r=0}^{n} r\binom{n}{r}^2 = nA_n - B_n$$

$$2B_n = nA_n \implies B_n = \frac{n}{2}A_n$$

Hence, statement (A) is true.

• Analysis for C_n:

Using the identity $r\binom{n}{r} = n\binom{n-1}{r-1}$:

$$C_n = \sum_{r=1}^{n}\left[n\binom{n-1}{r-1}\right]^2 = n^2\sum_{r=1}^{n}\binom{n-1}{r-1}^2 = n^2\binom{2n-2}{n-1}$$

Hence, statement (B) is true.

• Analysis for the ratio $\frac{C_n}{B_n}$:

We know that:

$$\binom{2n-2}{n-1} = \frac{n}{2(2n-1)}\binom{2n}{n}$$

Substituting this into the expression for C_n:

$$C_n = n^2 \cdot \frac{n}{2(2n-1)}\binom{2n}{n} = \frac{n^3}{2(2n-1)}A_n$$

Now, computing the ratio:

$$\frac{C_n}{B_n} = \frac{\frac{n^3}{2(2n-1)}A_n}{\frac{n}{2}A_n} = \frac{n^2}{2n-1}$$

Hence, statement (C) is true.

• Verification for $n = 4$:

For $n = 3$:

$$A_3 = \binom{6}{3} = 20$$

$$B_3 = \frac{3}{2}A_3 = 30$$

$$A_3 + B_3 = 20 + 30 = 50 \neq 60$$

Hence, statement (D) is false.

Q11. Match the characterizations/expressions in List-I with their corresponding values in List-II.

List-I		List-II	
(A)	The coefficient of x^2 in the expansion of $(1 + x + 2x^2)^4$ is	(P)	6
(B)	The number of ways to arrange 5 identical red balls and 2 identical black balls in a row such that no two black balls are adjacent is	(Q)	14
(C)	The number of strictly increasing sequences of length 3 chosen from the set {1, 2, 3, 4, 5, 6, 7} is	(R)	15
(D)	The number of 4-digit numbers that can be formed using the digits 1, 2, 3, 4, 5 without repetition such that the number is divisible by 3 is	(S)	24
		(T)	35

Choose the correct option :

(A) $A \to Q, B \to R, C \to T, D \to S$ (B) $A \to P, B \to R, C \to S, D \to T$

(C) $A \to Q, B \to P, C \to T, D \to S$ (D) $A \to P, B \to Q, C \to R, D \to S$

Ans. A

Sol. For (A): Using the multinomial theorem, the general term in the expansion of $(1 + x + 2x^2)^4$ is given by :

$$\frac{4!}{a!b!c!}(1)^a(x)^b(2x^2)^c = \frac{4!}{a!b!c!}2^c x^{b+2c}$$

where $a + b + c = 4$ and $a, b, c \in N_0$.

For the coefficient of x^2, we require $b + 2c = 2$.

Case 1: $c = 0 \Rightarrow b = 2 \Rightarrow a = 2$. The term is $\frac{4!}{2!2!0!}2^0 = 6$.

Case 2: $c = 1 \Rightarrow b = 0 \Rightarrow a = 3$. The term is $\frac{4!}{3!0!1!}2^1 = 8$.

Total coefficient = 6 + 8 = 14. Thus, (A) $\to$ (Q).

For (B): First, arrange the 5 identical red balls in a row (1 way). This creates 6 available gaps to place the black balls : _ R_R_R_R_R_

To ensure no two black balls are adjacent, we select 2 gaps out of the 6 for the 2 identical black balls.

Number of ways = $\binom{6}{2} = 15$. Thus, (B) $\to$ (R).

For (C): Any unique selection of 3 distinct elements from the 7-element set {1, 2, 3, 4, 5, 6, 7} can be ordered in exactly one strictly increasing sequence.

Number of sequences = $\binom{7}{3} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$. Thus, (C) $\to$ (T).

For (D): The sum of the digits {1, 2, 3, 4, 5} is 15. For a 4-digit number to be divisible by 3, the sum of its digits must be a multiple of 3.

When excluding one digit d from the set, the remaining sum 15-d must be divisible by 3, which implies d must be a multiple of 3. The only such digit in the set is 3.

Therefore, the selected 4 digits must be $\{1, 2, 4, 5\}$.

The number of arrangements of these digits without repetition is 4! = 24. Thus, (D) $\rightarrow$ (S).

Q12. Match the conditions/expressions in List-I with their corresponding integer values in List-II.

List-I	List-II
(P) The number of ways to divide 6 distinct books into 3 equal groups is	(1) 5
(Q) A fair coin is tossed 4 times. If the probability of getting more heads than tails is p, then the value of $16p$ is	(2) 6
(R) A committee of 3 members is to be selected from 4 men and 3 women. If the probability that the committee contains at least 2 women is q, then the value of $35q$ is	(3) 12
(S) The number of ways to choose 2 non-consecutive integers from the set $\{1, 2, 3, 4, 5\}$ is	(4) 13
	(5) 15

(A) $\mathrm{P} \rightarrow 5; \mathrm{Q} \rightarrow 1; \mathrm{R} \rightarrow 4; \mathrm{S} \rightarrow 2$ (B) $\mathrm{P} \rightarrow 5; \mathrm{Q} \rightarrow 2; \mathrm{R} \rightarrow 4; \mathrm{S} \rightarrow 1$

(C) $\mathrm{P} \rightarrow 2; \mathrm{Q} \rightarrow 1; \mathrm{R} \rightarrow 3; \mathrm{S} \rightarrow 5$ (D) $\mathrm{P} \rightarrow 5; \mathrm{Q} \rightarrow 1; \mathrm{R} \rightarrow 3; \mathrm{S} \rightarrow 2$

Ans. A

Sol. For (P):

The number of ways to divide 6 distinct objects into 3 groups of equal size (2 items each) is given by:

$$\frac{6!}{(2!)^3 \times 3!} = \frac{720}{8 \times 6} = 15$$

Thus, $\mathrm{P} \rightarrow 5$.

For (Q):

The total number of outcomes when a coin is tossed 4 times is $2^4 = 16$.

The event "more heads than tails" corresponds to getting either 3 heads or 4 heads.

$$\text{Number of favorable outcomes} = \binom{4}{3} + \binom{4}{4} = 4 + 1 = 5$$

$$p = \frac{5}{16} \implies 16p = 5$$

Thus, $\mathrm{Q} \rightarrow 1$.

For (R):

The total number of ways to select a 3-member committee from 7 people (4 men and 3 women) is:

$$\text{Total outcomes} = \binom{7}{3} = 35$$

The committee contains at least 2 women if it has 2 women and 1 man, or 3 women:

$$\text{Favorable outcomes} = \binom{3}{2} \times \binom{4}{1} + \binom{3}{3} = 3 \times 4 + 1 = 13$$

$$q = \frac{13}{35} \implies 35q = 13$$

Thus, $\mathrm{R} \to 4$.

For (S):

The total number of ways to choose any 2 integers from $\{1, 2, 3, 4, 5\}$ is $\binom{5}{2} = 10$.

The consecutive pairs are $(1, 2), (2, 3), (3, 4)$, and $(4, 5)$, which equals 4 pairs.

$$\text{Number of non-consecutive pairs} = 10 - 4 = 6$$

Alternatively, using the formula $\binom{n-k+1}{k}$ for $n = 5, k = 2$:

$$\binom{5-2+1}{2} = \binom{4}{2} = 6$$

Thus, $\mathrm{S} \to 2$.

Combining all the matchings:

$$\mathrm{P} \to 5; \mathrm{Q} \to 1; \mathrm{R} \to 4; \mathrm{S} \to 2$$

Q13. Let $y = f(x)$ be a real-valued differentiable function satisfying the implicit relation $\ln(1 + y) + x^2 y + x = 0$ for all $x \in (-1, 1)$ with $f(0) = 0$. If $f'''(0)$ denotes the third-order derivative of $f(x)$ evaluated at $x = 0$, then the value of $f'''(0)$ is

Ans. 5

Sol. Using series expansion for $y = f(x)$ around $x = 0$:

$$y = a_1 x + a_2 x^2 + a_3 x^3 + \ldots$$

where $a_1 = f'(0)$, $a_2 = \dfrac{f''(0)}{2!}$, and $a_3 = \dfrac{f'''(0)}{3!}$.

Substituting this into the expansion for $\ln(1 + y)$:

$$\ln(1 + y) = y - \frac{y^2}{2} + \frac{y^3}{3} - \ldots$$

$$\ln(1 + y) = (a_1 x + a_2 x^2 + a_3 x^3) - \frac{1}{2}(a_1 x + a_2 x^2)^2 + \frac{1}{3}(a_1 x)^3 + \ldots$$

$$\ln(1 + y) = a_1 x + \left(a_2 - \frac{1}{2} a_1^2\right) x^2 + \left(a_3 - a_1 a_2 + \frac{1}{3} a_1^3\right) x^3 + \ldots$$

Now, substitute this back into the given implicit equation $\ln(1+y)+x^2y+x=0$:

$$(a_1x+a_2x^2+a_3x^3-\frac{1}{2}a_1^2x^2-a_1a_2x^3+\frac{1}{3}a_1^3x^3)+x^2(a_1x+\ldots)+x=0$$

Grouping the coefficients of like powers of x:

$$(a_1+1)x+\left(a_2-\frac{1}{2}a_1^2\right)x^2+\left(a_3-a_1a_2+\frac{1}{3}a_1^3+a_1\right)x^3+\cdots=0$$

Equating the coefficients to zero:

1. For the coefficient of x:

$$a_1+1=0 \implies a_1=-1$$

2. For the coefficient of x^2:

$$a_2-\frac{1}{2}a_1^2=0 \implies a_2=\frac{1}{2}(-1)^2=\frac{1}{2}$$

3. For the coefficient of x^3:

$$a_3-a_1a_2+\frac{1}{3}a_1^3+a_1=0$$

Substitute the values of a_1 and a_2:

$$a_3-(-1)\left(\frac{1}{2}\right)+\frac{1}{3}(-1)^3+(-1)=0$$

$$a_3+\frac{1}{2}-\frac{1}{3}-1=0 \implies a_3=\frac{5}{6}$$

Since $a_3=\dfrac{f'''(0)}{3!}=\dfrac{f'''(0)}{6}$, we get:

$$\frac{f'''(0)}{6}=\frac{5}{6} \implies f'''(0)=5$$

Q14. Let $f:[-2,2]\to\mathbb{R}$ be a continuous function defined by:

$$f(x)=\begin{cases}\frac{a(1-\cos 2x)}{x^2}-2x & \text{for } -2\le x<0\\ b & \text{for } x=0\\ \lim_{t\to 0}\frac{\sin(x+t)-\sin x}{t}+x^2-3 & \text{for } 0<x\le 2\end{cases}$$

If N represents the total number of real roots of $f(x)=0$ in the interval $[-2,2]$, then find the value of $|a|+|b|+N$.

Ans. 5

Sol. To ensure continuity at $x=0$, the left-hand limit, right-hand limit, and the value of the function at $x=0$ must be equal.

Evaluating the left-hand limit ($x\to 0^-$):

$$\lim_{x\to 0^-}f(x)=\lim_{x\to 0^-}\left[\frac{a(2\sin^2 x)}{x^2}-2x\right]=2a$$

Evaluating the right-hand limit ($x \to 0^+$):

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \left[\cos x + x^2 - 3\right] = 1 + 0 - 3 = -2$$

Equating the limits for continuity at $x = 0$:

$$2a = -2 \implies a = -1$$

$$b = f(0) = -2$$

Thus, $|a| = 1$ and $|b| = 2$.
Now, we determine the total number of real roots N of $f(x) = 0$ across the defined intervals:
• Case I: $x \in (0, 2]$

$$f(x) = \cos x + x^2 - 3$$

Since $f(0^+) = -2 < 0$ and $f(2) = \cos 2 + 1 > 0$, the Intermediate Value Theorem (IVT) guarantees at least one root in $(0, 2)$.
Differentiating the function gives $f'(x) = 2x - \sin x$. Since $2x > \sin x$ for all $x > 0$, $f'(x) > 0$. The function is strictly increasing, yielding exactly 1 real root.
• Case II: $x = 0$

$$f(0) = -2 \neq 0 \implies \text{No root at } x = 0.$$

• Case III: $x \in [-2, 0)$

$$f(x) = -\frac{1 - \cos 2x}{x^2} - 2x = -2\frac{\sin^2 x}{x^2} - 2x$$

Since $f(-2) = 4 - \dfrac{\sin^2 2}{2} > 0$ and $\lim\limits_{x \to 0^-} f(x) = -2 < 0$, IVT guarantees at least one root in $(-2, 0)$
.
Differentiating the function gives:

$$f'(x) = -\frac{4}{x^3}\sin x(x\cos x - \sin x) - 2$$

For $x \in (-2, 0)$, we have $x^3 < 0$, $\sin x < 0$, and $(x \cos x - \sin x) > 0$. This confirms the first term is negative, so $f'(x) < 0$. The function is strictly decreasing, yielding exactly 1 real root.
Therefore, the total number of real roots is $N = 1 + 1 = 2$.
The final required value is:

$$|a| + |b| + N = 1 + 2 + 2 = 5$$

Q15. Let $L=\lim\limits_{n\to\infty}\sum\limits_{k=1}^{n}\dfrac{8k^3+\left[k\ln\left(1+\frac{k}{n}\right)\right]}{n^4+k^2\sin k}$, where $\lfloor x\rfloor$ denotes the greatest integer less than or equal to x. Then the value of L is equal to _____.

Ans. 2

Sol. By the definition of the greatest integer function, we have the following inequality for the numerator for each $k\in\{1,2,\ldots,n\}$:

$$8k^3+k\ln\left(1+\frac{k}{n}\right)-1<8k^3+\left[k\ln\left(1+\frac{k}{n}\right)\right]\le 8k^3+k\ln\left(1+\frac{k}{n}\right)$$

For the denominator, since $-1\le\sin k\le 1$ and $1\le k\le n$, we have:

$$n^4-n^2\le n^4+k^2\sin k\le n^4+n^2$$

Combining these inequalities, we get the lower and upper bounds for the general term of the summation:

$$\frac{8k^3+k\ln\left(1+\frac{k}{n}\right)-1}{n^4+n^2}\le\frac{8k^3+\left[k\ln\left(1+\frac{k}{n}\right)\right]}{n^4+k^2\sin k}\le\frac{8k^3+k\ln\left(1+\frac{k}{n}\right)}{n^4-n^2}$$

Summing from $k=1$ to n:

$$\frac{\sum\limits_{k=1}^{n}8k^3+\sum\limits_{k=1}^{n}k\ln\left(1+\frac{k}{n}\right)-n}{n^4+n^2}\le\sum_{k=1}^{n}\frac{8k^3+\left[k\ln\left(1+\frac{k}{n}\right)\right]}{n^4+k^2\sin k}\le\frac{\sum\limits_{k=1}^{n}8k^3+\sum\limits_{k=1}^{n}k\ln\left(1+\frac{k}{n}\right)}{n^4-n^2}$$

We evaluate the order of the terms in the numerator as $n\to\infty$:

1. $\displaystyle\sum_{k=1}^{n}8k^3=8\cdot\frac{n^2(n+1)^2}{4}=2n^4+4n^3+2n^2$
2. $\displaystyle\sum_{k=1}^{n}k\ln\left(1+\frac{k}{n}\right)\le\sum_{k=1}^{n}n\ln 2=n^2\ln 2$, which is of order $O(n^2)$.
3. The remaining constant and linear terms are of lower order.

Taking the limit as $n\to\infty$ for the lower bound:

$$\lim_{n\to\infty}\frac{2n^4+4n^3+2n^2+O(n^2\ln 2)-n}{n^4+n^2}=\lim_{n\to\infty}\frac{2n^4}{n^4}=2$$

Taking the limit as $n\to\infty$ for the upper bound:

$$\lim_{n\to\infty}\frac{2n^4+4n^3+2n^2+O(n^2\ln 2)}{n^4-n^2}=\lim_{n\to\infty}\frac{2n^4}{n^4}=2$$

By the Sandwich (Squeeze) Theorem, the limit L must be equal to 2.

Q16. Consider a bijective function $f:\mathbb{R}\to\mathbb{R}$ defined by

$$f(x)=\ln\left(x+\sqrt{1+x^2}\right)+e^x+x$$

If $g(x)$ denotes the inverse function of $f(x)$, then the value of

$$81g'''(1) + 54g''(1) + 18g'(1)$$

\nis equal to _____.

Ans. 5

Sol. Given, $f(x) = \ln\left(x + \sqrt{1+x^2}\right) + e^x + x$.

Since $f(0) = 1$, we have $g(1) = 0$, where $g(x) = f^{-1}(x)$.

Differentiating $f(x)$ with respect to x:

$$f'(x) = \frac{1}{\sqrt{1+x^2}} + e^x + 1 \implies f'(0) = 3$$

Differentiating again:

$$f''(x) = -\frac{x}{(1+x^2)^{3/2}} + e^x \implies f''(0) = 1$$

Differentiating a third time:

$$f'''(x) = -\frac{(1+x^2)^{3/2} - 3x^2\sqrt{1+x^2}}{(1+x^2)^3} + e^x \implies f'''(0) = -1 + 1 = 0$$

Using the properties of the derivative of an inverse function:

$$g'(x) = \frac{1}{f'(g(x))} \implies g'(1) = \frac{1}{f'(0)} = \frac{1}{3}$$

For the second derivative:

$$g''(x) = -\frac{f''(g(x))}{(f'(g(x)))^3} \implies g''(1) = -\frac{f''(0)}{(f'(0))^3} = -\frac{1}{27}$$

For the third derivative:

$$g'''(x) = -\frac{f'''(g(x))}{(f'(g(x)))^4} + \frac{3(f''(g(x)))^2}{(f'(g(x)))^5}$$

$$\implies g'''(1) = -\frac{0}{3^4} + \frac{3(1)^2}{3^5} = \frac{1}{81}$$

Substituting these values into the required expression:

$$81g'''(1) + 54g''(1) + 18g'(1) = 81\left(\frac{1}{81}\right) + 54\left(-\frac{1}{27}\right) + 18\left(\frac{1}{3}\right)$$

$$= 1 - 2 + 6 = 5$$

Q17. Let $f : (-1, 1) \to \mathbb{R}$ be a function defined by

$$f(x) = \lim_{n \to \infty} \left(\sqrt[3]{\cos\left(\frac{x\sqrt{6}}{\sqrt{n}}\right)} + \frac{1}{n(1-x^2)} \right)^n$$

If the value of the limit

$$\lim_{x \to 0} \frac{2 \ln f(x) - 2 - \ln(1 + 2x^4 + 2x^6)}{x^8}$$

is equal to L, then the value of L is _____.

Ans. 4

Sol. Using the Taylor series expansion for $\cos(t)$ as $n \to \infty$:

$$\cos\left(\frac{x\sqrt{6}}{\sqrt{n}}\right) = 1 - \frac{3x^2}{n} + \frac{3x^4}{2n^2} + O\left(\frac{1}{n^3}\right)$$

Using the binomial expansion $(1 - z)^{1/3} = 1 - \frac{1}{3}z - \frac{1}{9}z^2 - \ldots$:

$$\sqrt[3]{\cos\left(\frac{x\sqrt{6}}{\sqrt{n}}\right)} = 1 - \frac{x^2}{n} - \frac{x^4}{2n^2} + O\left(\frac{1}{n^3}\right)$$

The second term inside the limit can be expanded as:

$$\frac{1}{n(1-x^2)} = \frac{1}{n}\left(1 + x^2 + x^4 + x^6 + \ldots\right)$$

Adding these two expressions gives the base of the limit:

$$\text{Base} = 1 + \frac{1}{n}\left(1 + x^4 + x^6 + x^8 + \ldots\right) - \frac{x^4}{2n^2} + O\left(\frac{1}{n^3}\right)$$

Using the standard limit $\lim_{n \to \infty} \left(1 + \frac{A}{n} + \frac{B}{n^2}\right)^n = e^A$:

$$f(x) = e^{1 + x^4 + x^6 + x^8 + \ldots}$$

Taking the natural logarithm on both sides:

$$\ln f(x) = 1 + x^4 + x^6 + x^8 + x^{10} + \ldots$$

Thus, we have:

$$2 \ln f(x) - 2 = 2x^4 + 2x^6 + 2x^8 + 2x^{10} + \ldots$$

Next, expand the logarithmic term in the numerator using $\ln(1 + z) = z - \frac{z^2}{2} + \ldots$ where

$z = 2x^4 + 2x^6$:

$$\ln(1 + 2x^4 + 2x^6) = (2x^4 + 2x^6) - \frac{(2x^4 + 2x^6)^2}{2} + O(x^{12})$$

$$\ln(1 + 2x^4 + 2x^6) = 2x^4 + 2x^6 - 2x^8 - 4x^{10} + O(x^{12})$$

Now, substitute these expansions into the target limit:

$$L = \lim_{x \to 0} \frac{(2x^4 + 2x^6 + 2x^8 + 2x^{10} + \ldots) - (2x^4 + 2x^6 - 2x^8 - 4x^{10} + \ldots)}{x^8}$$

$$L = \lim_{x \to 0} \frac{4x^8 + 6x^{10} + \ldots}{x^8} = 4$$

Q18. Consider the following discrete frequency distribution for a variable x:

x	1	2	3	4	5
f	1	2	6	k	3

where k is a positive integer. It is given that the distribution has a unique mode. If the mean deviation of the distribution about its mode is 0.875, and the mean deviation about its mean is D, then the value of $64D$ is equal to _____.

Ans. 59

Sol. Since the distribution has a unique mode and k is an integer, the mode must be 3 with a maximum frequency of 6 (which requires $k < 6$).
The mean deviation about the mode (3) is given by:

$$\text{MD}_{\text{mode}} = \frac{\sum f_i |x_i - 3|}{\sum f_i}$$

$$\text{MD}_{\text{mode}} = \frac{1|1-3| + 2|2-3| + 6|3-3| + k|4-3| + 3|5-3|}{1 + 2 + 6 + k + 3}$$

$$\text{MD}_{\text{mode}} = \frac{2 + 2 + 0 + k + 6}{12 + k} = \frac{10 + k}{12 + k}$$

We are given that $\text{MD}_{\text{mode}} = 0.875 = \frac{7}{8}$:

$$\frac{10 + k}{12 + k} = \frac{7}{8} \implies 80 + 8k = 84 + 7k \implies k = 4$$

Since $k = 4 < 6$, the unique mode is indeed 3.
Now, we find the mean ($\bar{x}$) of the distribution for $k = 4$:

$$\bar{x} = \frac{1(1) + 2(2) + 6(3) + 4(4) + 3(5)}{1 + 2 + 6 + 4 + 3} = \frac{1 + 4 + 18 + 16 + 15}{16} = \frac{54}{16} = 3.375$$

The mean deviation about the mean (D) is:

$$D = \frac{\sum f_i|x_i - 3.375|}{16}$$

$$\sum f_i|x_i - 3.375| = 1|1 - 3.375| + 2|2 - 3.375| + 6|3 - 3.375| + 4|4 - 3.375| + 3|5 - 3.375|$$

$$= 1(2.375) + 2(1.375) + 6(0.375) + 4(0.625) + 3(1.625)$$

$$= 2.375 + 2.75 + 2.25 + 2.5 + 4.875 = 14.75$$

Thus, we have:

$$D = \frac{14.75}{16} = \frac{59}{64}$$

Therefore:

$$64D = 59$$